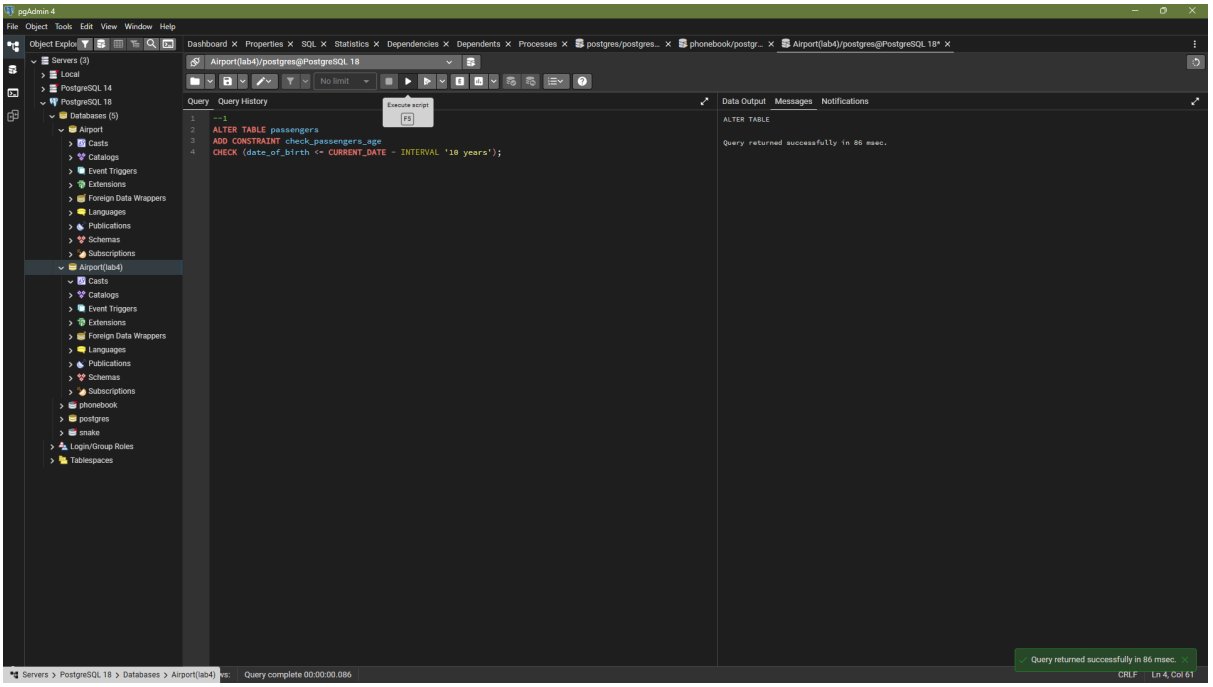
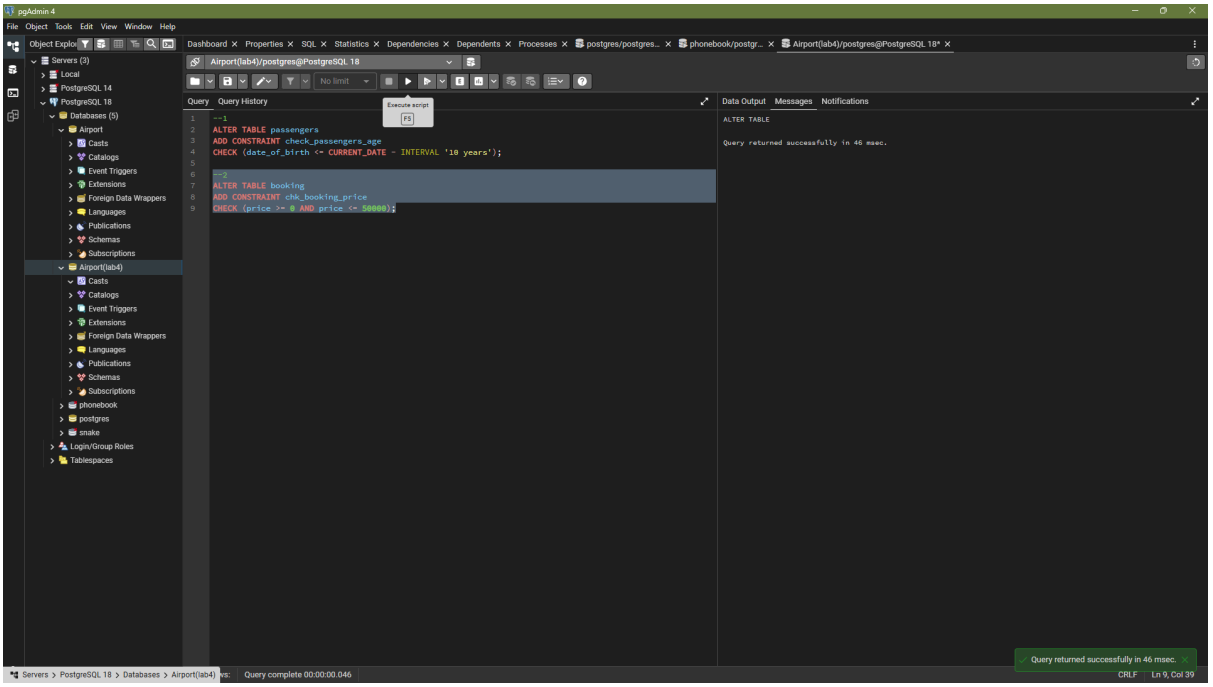
Lab 5

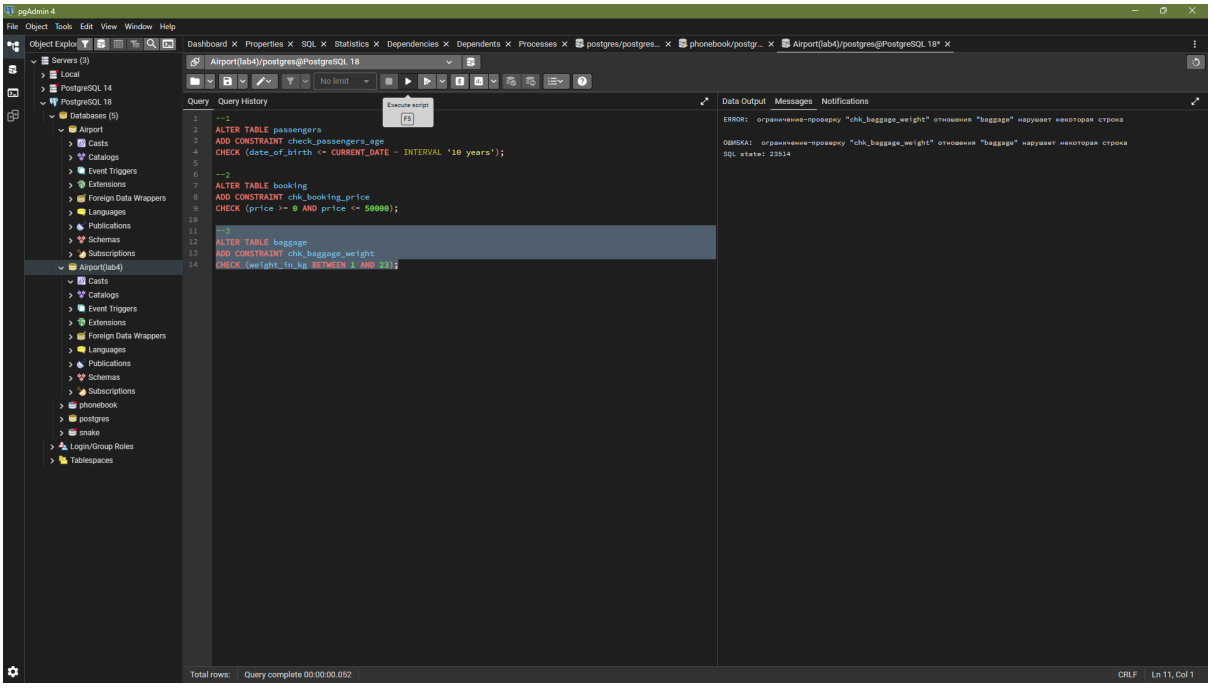
1.


pgAdmin 4
File Object Tools Edit View Window Help
Dashboard Properties SQL Statistics Dependencies Dependents Processes postgres/postgres... phonebook/postgr... Airport(lab4)/postgres@PostgreSQL 18*
Airport(lab4)/postgres@PostgreSQL 18
Query Query History
Execute script
F5
--1
ALTER TABLE passengers
ADD CONSTRAINT check_passengers_age
CHECK (date_of_birth <= CURRENT_DATE - INTERVAL '18 years');
Data Output Messages Notifications
ALTER TABLE
Query returned successfully in 86 msec.
Query returned successfully in 86 msec.
Query complete 00:00:00.086
CRLF Ln 4, Col 61

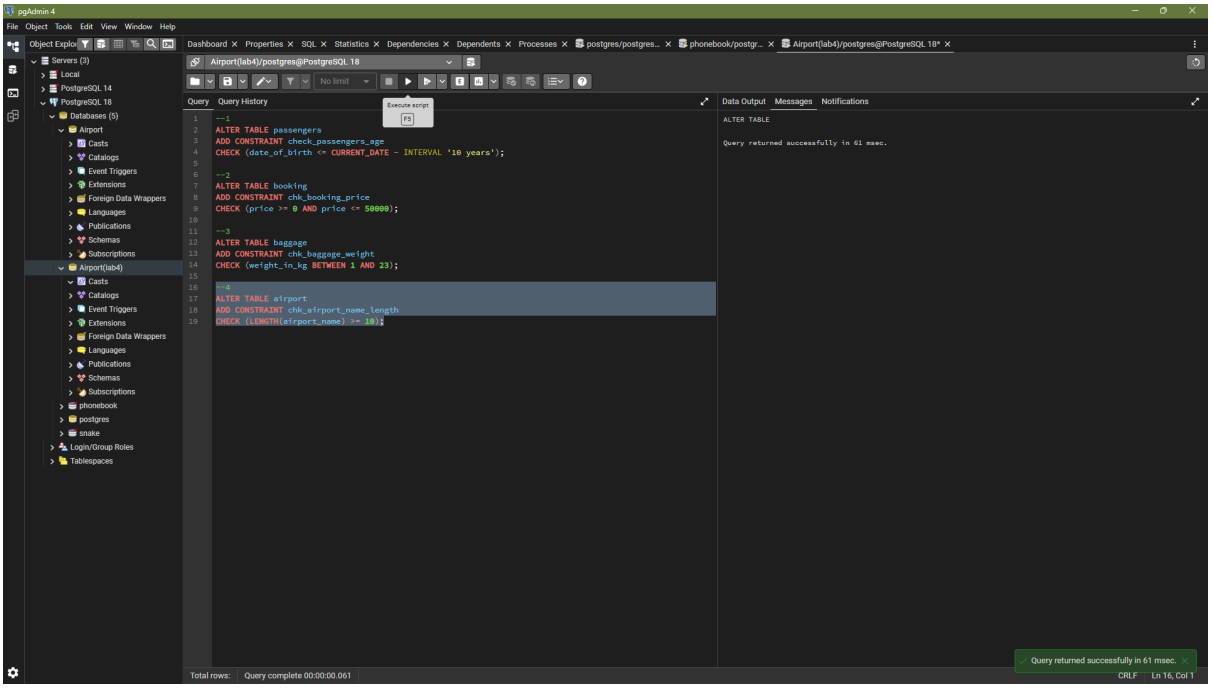
2.


pgAdmin 4
File Object Tools Edit View Window Help
Dashboard Properties SQL Statistics Dependencies Dependents Processes postgres/postgres... phonebook/postgr... Airport(lab4)/postgres@PostgreSQL 18*
Airport(lab4)/postgres@PostgreSQL 18
Query Query History
Execute script
F5
--1
ALTER TABLE passengers
ADD CONSTRAINT check_passengers_age
CHECK (date_of_birth <= CURRENT_DATE - INTERVAL '18 years');
--2
ALTER TABLE booking
ADD CONSTRAINT chk_booking_price
CHECK (price >= 0 AND price <= 50000);
Data Output Messages Notifications
ALTER TABLE
Query returned successfully in 46 msec.
Query returned successfully in 46 msec.
Query complete 00:00:00.046
CRLF Ln 9, Col 39

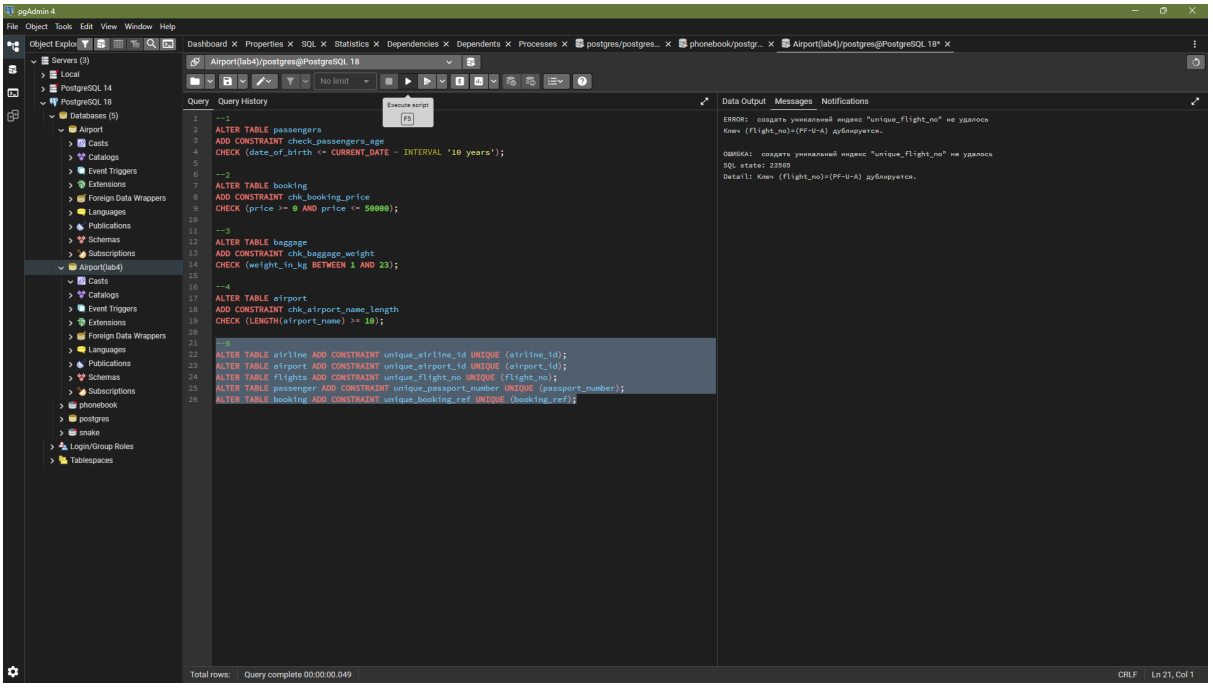
3.


pgAdmin 4
File Object Tools Edit View Window Help
Airport(lab4)/postgres@PostgreSQL 18
--1
ALTER TABLE passengers
ADD CONSTRAINT check_passengers_age
CHECK (date_of_birth <= CURRENT_DATE - INTERVAL '10 years');
--2
ALTER TABLE booking
ADD CONSTRAINT chk_booking_price
CHECK (price >= 0 AND price <= 50000);
--3
ALTER TABLE baggage
ADD CONSTRAINT chk_baggage_weight
CHECK (weight_in_kg BETWEEN 1 AND 23);
ERROR: ограничение-проверку "chk_baggage_weight" отношения "baggage" нарушает некоторая строка
ОШИБКА: ограничение-проверку "chk_baggage_weight" отношения "baggage" нарушает некоторая строка
SQL state: 23514
Total rows:
Query complete 00:00:00.052
CRLF
Ln 11, Col 1

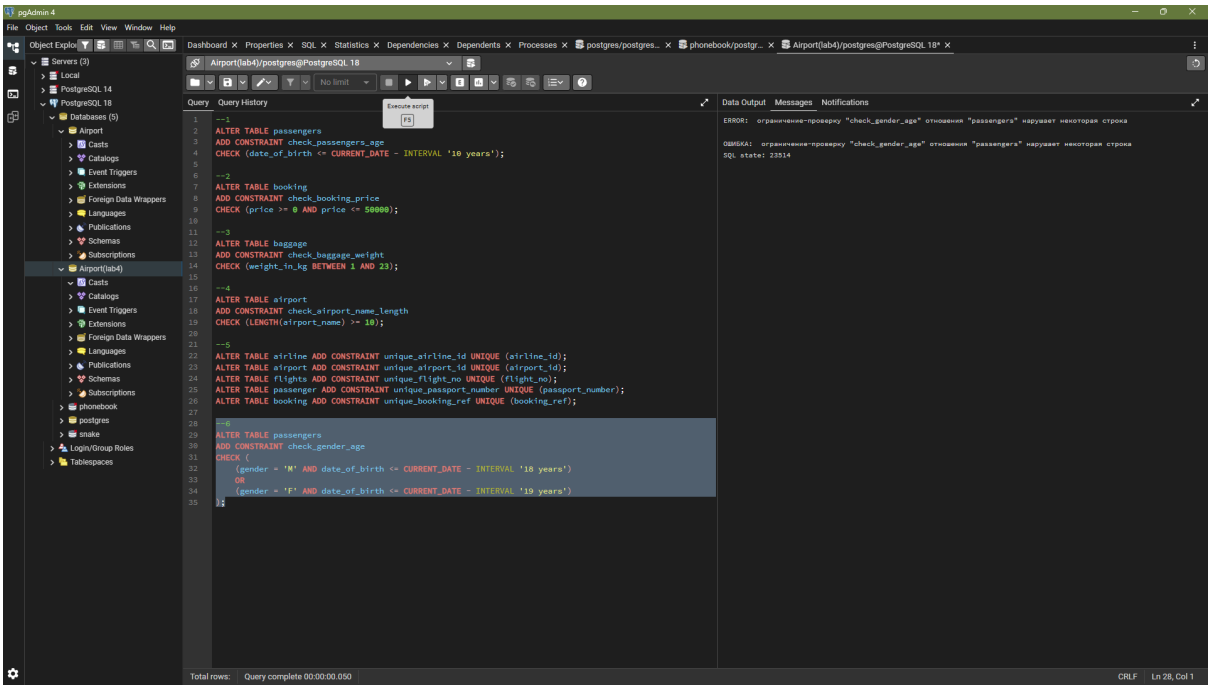
4.


pgAdmin 4
File Object Tools Edit View Window Help
Airport(lab4)/postgres@PostgreSQL 18
--1
ALTER TABLE passengers
ADD CONSTRAINT check_passengers_age
CHECK (date_of_birth <= CURRENT_DATE - INTERVAL '10 years');
--2
ALTER TABLE booking
ADD CONSTRAINT chk_booking_price
CHECK (price >= 0 AND price <= 50000);
--3
ALTER TABLE baggage
ADD CONSTRAINT chk_baggage_weight
CHECK (weight_in_kg BETWEEN 1 AND 23);
--4
ALTER TABLE airport
ADD CONSTRAINT chk_airport_name_length
CHECK (LENGTH(airport_name) >= 10);
ALTER TABLE
Query returned successfully in 61 msec.
Query returned successfully in 61 msec.
Total rows:
Query complete 00:00:00.061
CRLF
Ln 16, Col 1

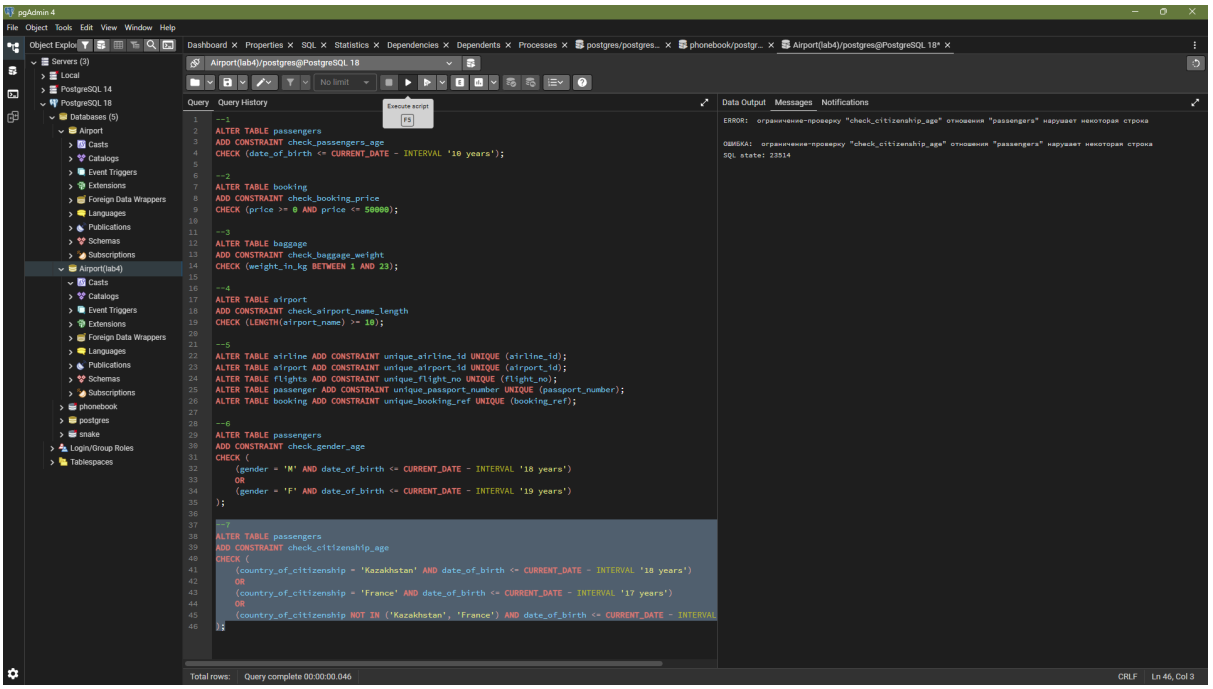
5.


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
Servers (3)
Local
PostgreSQL 14
PostgreSQL 18
Databases (5)
Airport
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas
Subscriptions
Airport(lab4)
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
Languages
Publications
Schemas
Subscriptions
phonebook
postgres
snake
Login/Group Roles
Tablespaces
Dashboard
Properties
SQL
Statistics
Dependencies
Dependents
Processes
postgres/postgres...
phonebook/postgr...
Airport(lab4)/postgres@PostgreSQL 18*
Airport(lab4)/postgres@PostgreSQL 18
No limit
Execute script
F5
Query
Query History
--1
ALTER TABLE passengers
ADD CONSTRAINT check_passengers_age
CHECK (date_of_birth <= CURRENT_DATE - INTERVAL '10 years');
--2
ALTER TABLE booking
ADD CONSTRAINT chk_booking_price
CHECK (price >= 0 AND price <= 50000);
--3
ALTER TABLE baggage
ADD CONSTRAINT chk_baggage_weight
CHECK (weight_in_kg BETWEEN 1 AND 23);
--4
ALTER TABLE airport
ADD CONSTRAINT chk_airport_name_length
CHECK (LENGTH(airport_name) >= 10);
--6
ALTER TABLE airline ADD CONSTRAINT unique_airline_id UNIQUE (airline_id);
ALTER TABLE airport ADD CONSTRAINT unique_airport_id UNIQUE (airport_id);
ALTER TABLE flights ADD CONSTRAINT unique_flight_no UNIQUE (flight_no);
ALTER TABLE passenger ADD CONSTRAINT unique_passport_number UNIQUE (passport_number);
ALTER TABLE booking ADD CONSTRAINT unique_booking_ref UNIQUE (booking_ref);
Data Output
Messages
Notifications
ERROR: создать уникальный индекс "unique_flight_no" не удалось
Ключ (flight_no)=(PF-U-A) дублируется.
ОШИБКА: создать уникальный индекс "unique_flight_no" не удалось
SQL state: 23505
Detail: Ключ (flight_no)=(PF-U-A) дублируется.
Total rows:
Query complete 00:00:00.049
CRLF
Ln 21, Col 1

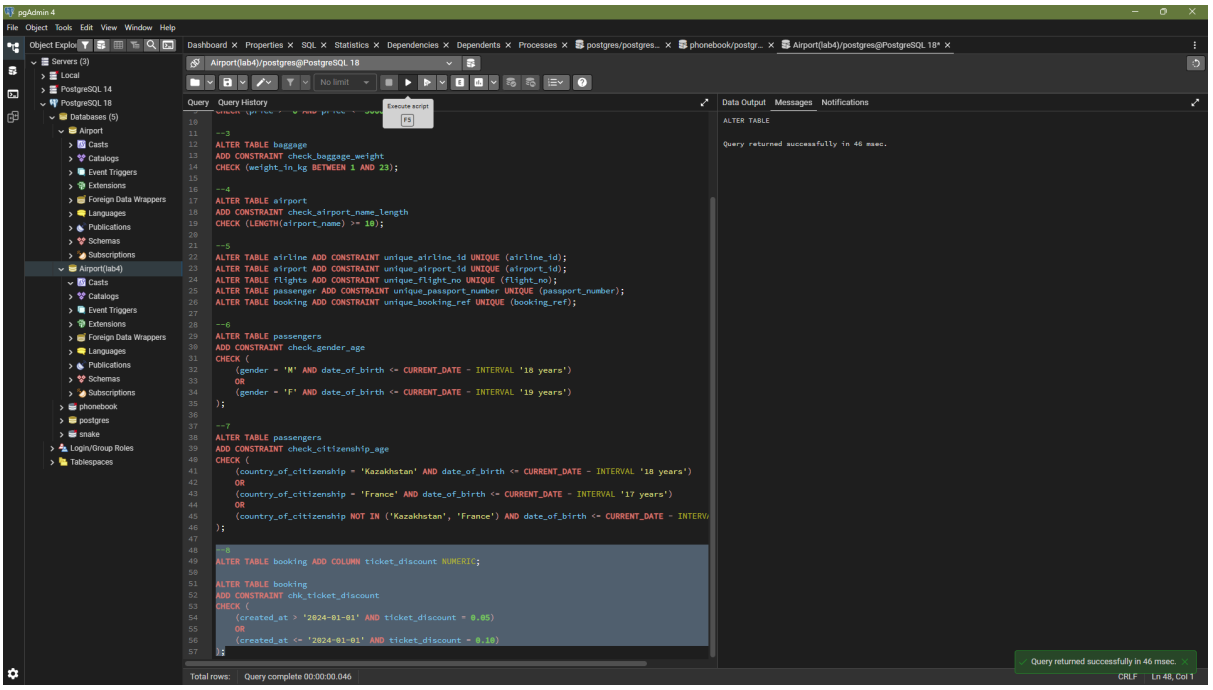
6.


--1
ALTER TABLE passengers
ADD CONSTRAINT check_passengers_age
CHECK (date_of_birth <= CURRENT_DATE - INTERVAL '10 years');
--2
ALTER TABLE booking
ADD CONSTRAINT check_booking_price
CHECK (price >= 0 AND price <= 50000);
--3
ALTER TABLE baggage
ADD CONSTRAINT check_baggage_weight
CHECK (weight_in_kg BETWEEN 1 AND 23);
--4
ALTER TABLE airport
ADD CONSTRAINT check_airport_name_length
CHECK (LENGTH(airport_name) >= 10);
--5
ALTER TABLE airline ADD CONSTRAINT unique_airline_id UNIQUE (airline_id);
ALTER TABLE airport ADD CONSTRAINT unique_airport_id UNIQUE (airport_id);
ALTER TABLE flights ADD CONSTRAINT unique_flight_no UNIQUE (flight_no);
ALTER TABLE passenger ADD CONSTRAINT unique_passport_number UNIQUE (passport_number);
ALTER TABLE booking ADD CONSTRAINT unique_booking_ref UNIQUE (booking_ref);
--6
ALTER TABLE passengers
ADD CONSTRAINT check_gender_age
CHECK (
(gender = 'M' AND date_of_birth <= CURRENT_DATE - INTERVAL '18 years')
OR
(gender = 'F' AND date_of_birth <= CURRENT_DATE - INTERVAL '19 years')
);
ERROR: ограничение-проверку "check_gender_age" отношения "passengers" нарушает некоторая строка
ОШИБКА: ограничение-проверку "check_gender_age" отношения "passengers" нарушает некоторая строка
SQL state: 23514

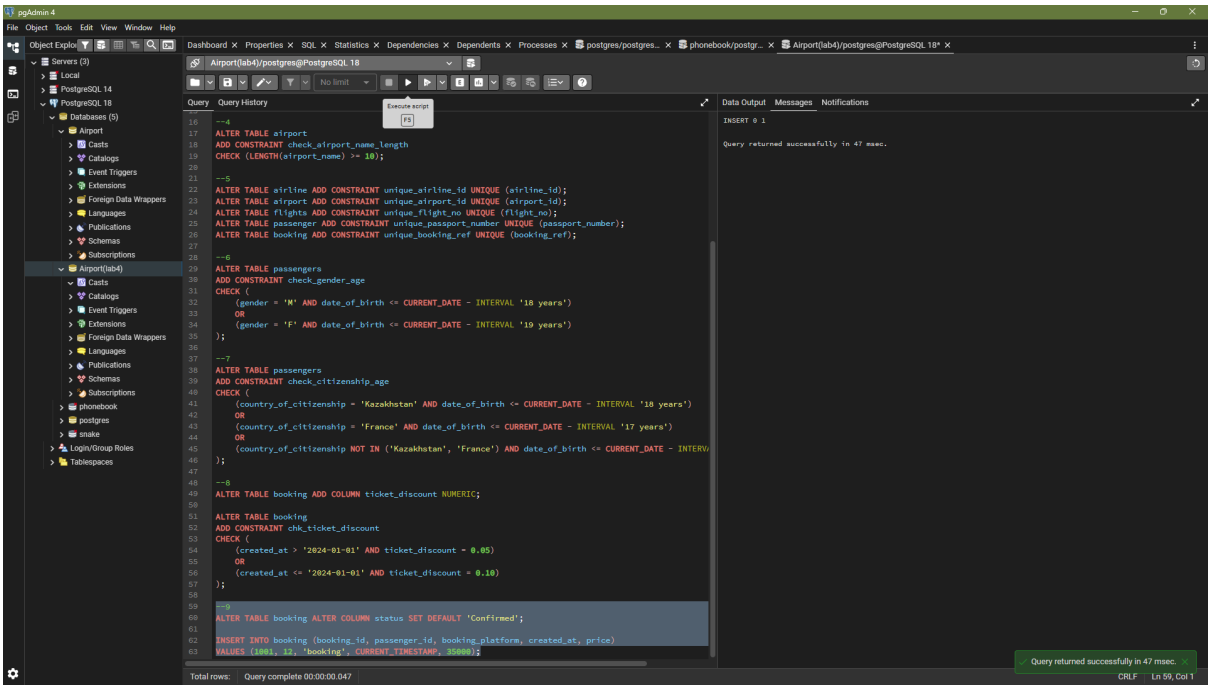
7.


--1
ALTER TABLE passengers
ADD CONSTRAINT check_passengers_age
CHECK (date_of_birth <= CURRENT_DATE - INTERVAL '10 years');
--2
ALTER TABLE booking
ADD CONSTRAINT check_booking_price
CHECK (price >= 0 AND price <= 50000);
--3
ALTER TABLE baggage
ADD CONSTRAINT check_baggage_weight
CHECK (weight_in_kg BETWEEN 1 AND 23);
--4
ALTER TABLE airport
ADD CONSTRAINT check_airport_name_length
CHECK (LENGTH(airport_name) >= 10);
--5
ALTER TABLE airline ADD CONSTRAINT unique_airline_id UNIQUE (airline_id);
ALTER TABLE airport ADD CONSTRAINT unique_airport_id UNIQUE (airport_id);
ALTER TABLE flights ADD CONSTRAINT unique_flight_no UNIQUE (flight_no);
ALTER TABLE passenger ADD CONSTRAINT unique_passport_number UNIQUE (passport_number);
ALTER TABLE booking ADD CONSTRAINT unique_booking_ref UNIQUE (booking_ref);
--6
ALTER TABLE passengers
ADD CONSTRAINT check_gender_age
CHECK (
(gender = 'M' AND date_of_birth <= CURRENT_DATE - INTERVAL '18 years')
OR
(gender = 'F' AND date_of_birth <= CURRENT_DATE - INTERVAL '19 years')
);
--7
ALTER TABLE passengers
ADD CONSTRAINT check_citizenship_age
CHECK (
(country_of_citizenship = 'Kazakhstan' AND date_of_birth <= CURRENT_DATE - INTERVAL '18 years')
OR
(country_of_citizenship = 'France' AND date_of_birth <= CURRENT_DATE - INTERVAL '17 years')
OR
(country_of_citizenship NOT IN ('Kazakhstan', 'France') AND date_of_birth <= CURRENT_DATE - INTERVAL
);
ERROR: ограничение-проверку "check_citizenship_age" отношения "passengers" нарушает некоторая строка
ОШИБКА: ограничение-проверку "check_citizenship_age" отношения "passengers" нарушает некоторая строка
SQL state: 23514

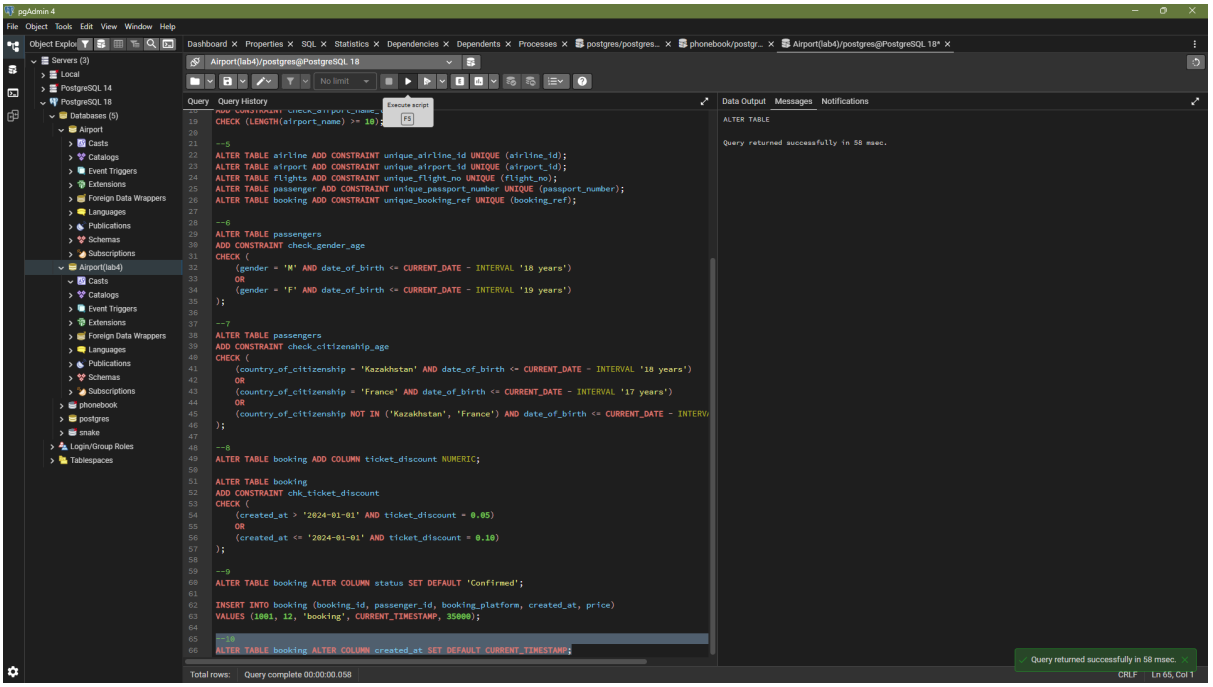
8.


pgAdmin 4
File Object Tools Edit View Window Help
Dashboard Properties SQL Statistics Dependencies Dependents Processes postgres/postgres... phonebook/postgr... Airport(lab4)/postgres@PostgreSQL 18*
Airport(lab4)/postgres@PostgreSQL 18
Query Query History
Data Output Messages Notifications
ALTER TABLE
Query returned successfully in 46 msec.
--3
ALTER TABLE baggage
ADD CONSTRAINT check_baggage_weight
CHECK (weight_in_kg BETWEEN 1 AND 23);
--4
ALTER TABLE airport
ADD CONSTRAINT check_airport_name_length
CHECK (LENGTH(airport_name) >= 10);
--5
ALTER TABLE airline ADD CONSTRAINT unique_airline_id UNIQUE (airline_id);
ALTER TABLE airport ADD CONSTRAINT unique_airport_id UNIQUE (airport_id);
ALTER TABLE flights ADD CONSTRAINT unique_flight_no UNIQUE (flight_no);
ALTER TABLE passenger ADD CONSTRAINT unique_passport_number UNIQUE (passport_number);
ALTER TABLE booking ADD CONSTRAINT unique_booking_ref UNIQUE (booking_ref);
--6
ALTER TABLE passengers
ADD CONSTRAINT check_gender_age
CHECK (
(gender = 'M' AND date_of_birth <= CURRENT_DATE - INTERVAL '16 years')
OR
(gender = 'F' AND date_of_birth <= CURRENT_DATE - INTERVAL '19 years')
);
--7
ALTER TABLE passengers
ADD CONSTRAINT check_citizenship_age
CHECK (
(country_of_citizenship = 'Kazakhstan' AND date_of_birth <= CURRENT_DATE - INTERVAL '18 years')
OR
(country_of_citizenship = 'France' AND date_of_birth <= CURRENT_DATE - INTERVAL '17 years')
OR
(country_of_citizenship NOT IN ('Kazakhstan', 'France') AND date_of_birth <= CURRENT_DATE - INTERV
);
--8
ALTER TABLE booking ADD COLUMN ticket_discount NUMERIC;
ALTER TABLE booking
ADD CONSTRAINT chk_ticket_discount
CHECK (
(created_at > '2024-01-01' AND ticket_discount = 0.05)
OR
(created_at <= '2024-01-01' AND ticket_discount = 0.10)
);
Query returned successfully in 46 msec.
Total rows: Query complete 00:00:00.046
CRLF Ln 48, Col 1

9.


pgAdmin 4
File Object Tools Edit View Window Help
Dashboard Properties SQL Statistics Dependencies Dependents Processes postgres/postgres... phonebook/postgr... Airport(lab4)/postgres@PostgreSQL 18*
Airport(lab4)/postgres@PostgreSQL 18
Query Query History
Data Output Messages Notifications
INSERT 0 1
Query returned successfully in 47 msec.
--4
ALTER TABLE airport
ADD CONSTRAINT check_airport_name_length
CHECK (LENGTH(airport_name) >= 10);
--5
ALTER TABLE airline ADD CONSTRAINT unique_airline_id UNIQUE (airline_id);
ALTER TABLE airport ADD CONSTRAINT unique_airport_id UNIQUE (airport_id);
ALTER TABLE flights ADD CONSTRAINT unique_flight_no UNIQUE (flight_no);
ALTER TABLE passenger ADD CONSTRAINT unique_passport_number UNIQUE (passport_number);
ALTER TABLE booking ADD CONSTRAINT unique_booking_ref UNIQUE (booking_ref);
--6
ALTER TABLE passengers
ADD CONSTRAINT check_gender_age
CHECK (
(gender = 'M' AND date_of_birth <= CURRENT_DATE - INTERVAL '16 years')
OR
(gender = 'F' AND date_of_birth <= CURRENT_DATE - INTERVAL '19 years')
);
--7
ALTER TABLE passengers
ADD CONSTRAINT check_citizenship_age
CHECK (
(country_of_citizenship = 'Kazakhstan' AND date_of_birth <= CURRENT_DATE - INTERVAL '18 years')
OR
(country_of_citizenship = 'France' AND date_of_birth <= CURRENT_DATE - INTERVAL '17 years')
OR
(country_of_citizenship NOT IN ('Kazakhstan', 'France') AND date_of_birth <= CURRENT_DATE - INTERV
);
--8
ALTER TABLE booking ADD COLUMN ticket_discount NUMERIC;
ALTER TABLE booking
ADD CONSTRAINT chk_ticket_discount
CHECK (
(created_at > '2024-01-01' AND ticket_discount = 0.05)
OR
(created_at <= '2024-01-01' AND ticket_discount = 0.10)
);
--9
ALTER TABLE booking ALTER COLUMN status SET DEFAULT 'Confirmed';
INSERT INTO booking (booking_id, passenger_id, booking_platform, created_at, price)
VALUES (1001, 12, 'booking', CURRENT_TIMESTAMP, 35000);
Query returned successfully in 47 msec.
Total rows: Query complete 00:00:00.047
CRLF Ln 59, Col 1

10.


pgAdmin 4
File Object Tools Edit View Window Help
Dashboard Properties SQL Statistics Dependencies Dependents Processes
Airport(lab4)/postgres@PostgreSQL 18
Query Query History
Execute script
F5
Data Output Messages Notifications
ALTER TABLE
Query returned successfully in 58 msec.
Total rows: Query complete 00:00:00.058
CRLF Ln 65, Col 1